Lab 07: Form, Value and Light Study

Level: Intermediate Suggested time: 75 minutes Platform: Complete the core brief in either Procreate or Sketchbook Studio; use the comparison notes to transfer the workflow. Alignment: K10 | A2 | A3 | LO2

Scenario

A ceramics maker needs a persuasive digital study of a new cup before a photo shoot.

Your role: Product visualisation artist Required deliverable: A rendered ceramic cup using primitives, a three-value plan and consistent light.

Learning goal

Apply Primitive construction | proportion | core/cast shadow | reflected light | material edges to a real design decision and produce evidence that meets explicit acceptance criteria.

Before you begin

- Use an iPad that supports your chosen app and a compatible stylus where available.
- Save an editable, layered master before exporting a flattened image.
- Use only owned, licensed or clearly permitted reference images and record their sources.
- Create a folder named Lab-07-<YourName> for working files and evidence.

Part A - Plan and construct

1. Photograph or select an owned reference of a plain ceramic cup under one clear light.
2. Create a square canvas and simplify the cup into cylinder, ellipses and handle forms.
3. Build a three-value thumbnail: light family, shadow family and cast shadow.
4. Refine plane turns with a core shadow, reflected light and controlled highlight.
5. Keep the cast-shadow direction and edge softness consistent with the light source.
6. Suggest ceramic through edge behaviour and highlight shape rather than stamped texture.
7. Export the greyscale study, colour study and a short light-direction diagram.

Part B - Procreate pathway

1. Use a simple opaque brush for the three-value block-in on separate form and shadow layers.
2. Use QuickShape ellipses and Transform Uniform to correct the cup structure.
3. Paint the core shadow with a soft-to-firm edge according to the form turn.
4. Create the cast shadow on its own layer below the cup; transform it to match the light direction.
5. Use a clipping layer for reflected light and highlights so they remain editable.
6. Check values with a greyscale view/adjustment on a duplicate.
7. Export the value plan and final colour study plus the layered master.

Part C - Sketchbook Studio pathway

1. Choose a blank or grid template and build the cup with Pencil, Monoline and the Ruler where useful.

2. Separate form, cast shadow, reflected light and highlight into layers.
3. Use brush opacity to control plane transitions; avoid global Soft Blur until the form reads.
4. Duplicate the form layer before filter experiments.
5. Use the precise eraser to recover the cup silhouette and clean cast-shadow edges.
6. Toggle Mono on a duplicate/export to validate the three-value grouping.
7. Export value and colour versions plus the layered sketch.

Evidence and acceptance checklist

[ ] Cup reads as a stable 3D form

[ ] Single light direction is consistent

[ ] Cast shadow supports placement

[ ] Material reads as ceramic without a texture stamp

[ ] Layered working file is saved and opens correctly.

[ ] Final export uses a clear filename and matches the intended dimensions.

[ ] At least one in-progress screenshot shows the technique in use.

[ ] Reference sources and design decisions are recorded.

Peer review questions

1. Which value transition creates roundness?
2. Where should the hardest edge appear?
3. How would matte clay change the highlight?

Debrief

Use the Observe → Interpret → Connect → Propose pattern:

1. Observe: State what is visible without praise or blame.
2. Interpret: Explain the likely effect on attention, meaning, emotion or use.
3. Connect: Link the effect to the brief, a design principle or the acceptance criteria.
4. Propose: Name one specific change and how you will check whether it works.

Record one decision you will keep, one change you will test, and one technique you can transfer to the other app.

Troubleshooting

- The line feels wobbly: Zoom out, slow the decision rather than the stroke, use shoulder movement, and increase stabilisation only enough to preserve intent.
- Colour leaks during fill: Close the contour, inspect gaps at 100%, or place colour on a separate layer and clean with the precise eraser.
- The file becomes hard to edit: Name or document layer roles, hide unused experiments, group related stages and duplicate before merging or filtering.
- The image looks muddy: Check value grouping, lower overlay opacity, remove competing texture and restore one clear focal contrast.

- Perspective feels wrong: Return to horizon and vanishing logic; correct the large box before windows, props or decoration.

Submission package

Submit Lab-07-<YourName>/ containing:

- layered working file (or the app's saved layered sketch);
- final PNG/JPEG/PDF as required by the brief;
- progress evidence/contact sheet;
- short design rationale answering the peer-review questions; and
- reference/provenance list.

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